EXPERIMENT 5

Document Question Answering using Retrieval-Augmented Generation (RAG)

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AIM

To demonstrate a document question-answering system by retrieving relevant context and producing an answer based on that context.

PROCEDURE

1. Prepare a small collection of sample documents.
2. Define a question about the documents.
3. Convert the documents and question into TF-IDF vectors.
4. Calculate similarity scores and retrieve the most relevant document.
5. Use the retrieved document as context for a sample answer.
6. Display the question, retrieved context, and answer.

PROGRAM

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

documents = [
    "Python is a popular programming language.",
    "Machine learning is a branch of AI that learns from data.",
    "SQL is used to store and manage information in databases."
]

question = "What is machine learning?"

vectorizer = TfidfVectorizer()
doc_vectors = vectorizer.fit_transform(documents)
question_vector = vectorizer.transform([question])

scores = cosine_similarity(question_vector, doc_vectors)[0]
best_index = scores.argmax()
retrieved_context = documents[best_index]

# Sample answer based on retrieved context
answer = (
    "Machine learning is a branch of AI that learns from data."
    if best_index == 1
    else "The retrieved document is: " + retrieved_context
)

print("Question:", question)
print("\nRetrieved Context:")
print(retrieved_context)
```

```
print("\nAnswer:")
print(answer)
```

OUTPUT

```
question = "What is machine learning?"

# Step 3: Retrieve the most relevant document
question_vector = vectorizer.transform([question])
scores = cosine_similarity(question_vector, document_vectors)

best_index = scores.argmax()
retrieved_document = documents[best_index]

# Step 4: Generate a simple answer from retrieved text
answer = retrieved_document

print("Question:", question)
print("\nRetrieved Document:", retrieved_document)
print("\nAnswer:", answer)
```

```
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.13/dist-packages (1.6.1)
Requirement already satisfied: numpy>=1.19.5 in /usr/local/lib/python3.13/dist-packages (from scikit-
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.13/dist-packages (from scikit-le
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.13/dist-packages (from scikit-
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.13/dist-packages (from s
Requirement already satisfied: cloudpickle>=3.0 in /usr/local/lib/python3.13/dist-packages (from joblib
Question: What is machine learning?

Retrieved Document: Machine learning allows computers to learn from data.

Answer: Machine learning allows computers to learn from data.
```

RESULT

Thus, a basic RAG workflow was demonstrated by retrieving relevant document context and producing an answer based on it.